EXERCISE 5

Finite Differences

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We solve the heat equation on $x \in [0,1]$ with $u(0) = u(1) = 0$, using finite differences on an equidistant grid with N interior points $x_j = j\,\Delta x$, $\Delta x = 1/(N+1)$.

1 Stationary heat equation $-u'' = f$ with $f = \pi^2 \sin \pi x$, exact solution $u = \sin \pi x$.

- **a)** Discretize u'' with the central 3-point stencil, set up the matrix A of $-\frac{d^2}{dx^2}$ as a sparse matrix (`scipy.sparse`) and solve $A\mathbf{u} = \mathbf{f}$.
  How can you check A before you solve anything?
- **b)** Plot the maximum error over Δx for $N = 3, 7, 15, \ldots, 10^6$.
  Which slope do you expect, which do you get? Below which Δx does the error grow again, and why?
- **c)** Try different discretizations of u'' (e.g. the higher-order stencils from the lecture) and compare them.
  Which orders do you get? What do you do close to the boundary?
- **d)** Replace the right boundary condition by $u'(1) = 0$, exact solution $u = \sin \frac{\pi x}{2}$ for $f = \frac{\pi^2}{4} \sin \frac{\pi x}{2}$.
  Which slope do you get? How does the discretization of the boundary condition affect the whole solution?
- **e)** Extend to 2D: $-\Delta u = 2\pi^2 \sin \pi x \sin \pi y$ on the unit square with $u = 0$ on the boundary, exact solution $u = \sin \pi x \sin \pi y$ (hint: `scipy.sparse.kron`).
  How do error and runtime scale with N?

2 Time-dependent heat equation $u_t = \alpha\, u_{xx}$ $(\alpha = 1)$ with $u(x,0) = \sin \pi x$, exact solution $u = e^{-\pi^2 t} \sin \pi x$.

- **a) Method of lines.** Discretize in space with A from 1 and solve the ODE system $\dot{\mathbf{u}} = -A\mathbf{u}$ with explicit Euler, implicit Euler and Crank–Nicolson.
  Which parts of your code from Exercise 3 can you reuse unchanged?
- **b) Accuracy.** Study the error at $t = 0.1$ in time ($N = 50$, varying Δt) and in space (small fixed Δt, varying N). Compare with the exact PDE solution and with the exact solution $e^{-\lambda_1 t} \sin \pi x_j$ of the ODE system (λ_1: smallest eigenvalue of A).
  Which slopes do you expect? Why does the error stop decreasing at some point?
- **c) Stability.** Calculate the eigenvalues λ_k and eigenvectors of A and plot some eigenvectors. Mark $z_k = -\lambda_k \Delta t$ in the stability regions of the ODE solvers from Exercise 3 (explicit Euler, Heun, RK4, implicit Euler) and of Crank–Nicolson. Determine the critical Δt for $N = 10, 20, 40, 80$ and compare with the von Neumann prediction.
  Which eigenvalue decides about stability? What does the solution look like just above the critical time step? Why are implicit Euler and Crank–Nicolson stable for every Δt?
- **d) Stiffness and non-smooth data.** Repeat 2b for the step $u(x,0) = 1$ for $0.25 < x < 0.75$, else 0 ($N = 99$), in particular for $\Delta t = 0.01$ ($r = 100$). Plot $|R(z)|$ of implicit Euler and Crank–Nicolson for $z \in [-1000, 0]$ together with e^z, and follow the amplitude of the highest eigenmode over time.
  Why is Crank–Nicolson, the best method in 2b, now worse than implicit Euler? Do the fast modes ever disappear? How many steps would explicit Euler need, and how does this change if you refine the grid?